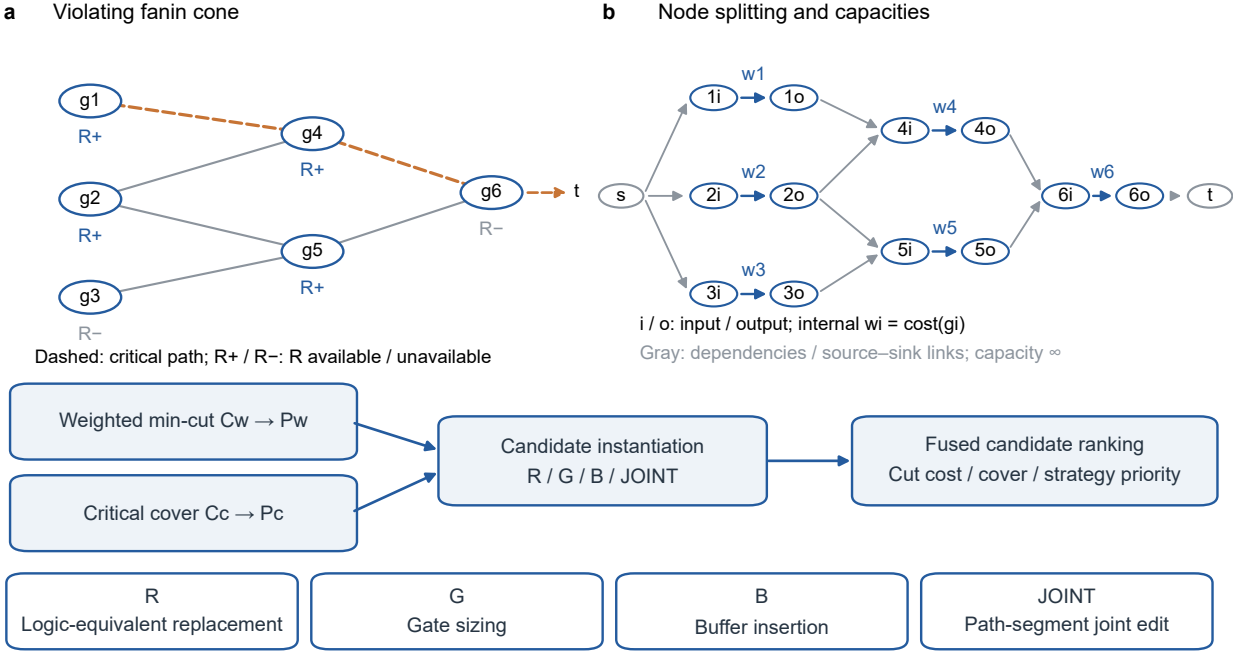

a
Violating fanin cone
g1
R+
g2
R+
g3
R−
g4
R+
g5
R+
g6
R−
t
Dashed: critical path; R+ / R−: R available / unavailable
b
Node splitting and capacities
s
1i
1o
w1
2i
2o
w2
3i
3o
w3
4i
4o
w4
5i
5o
w5
6i
6o
w6
t
i / o: input / output; internal wi = cost(gi)
Gray: dependencies / source–sink links; capacity ∞
Weighted min-cut Cw → Pw
Critical cover Cc → Pc
Candidate instantiation
R / G / B / JOINT
Fused candidate ranking
Cut cost / cover / strategy priority
R
Logic-equivalent replacement
G
Gate sizing
B
Buffer insertion
JOINT
Path-segment joint edit